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Procedia Computer Science 258 (2025) 1659–1668

Procedia Computer Science

www.elsevier.com/locate/procedia

International Conference on Machine Learning and Data Engineering

A Hybrid Deep Learning Malware Detection Model with The Fusion of VGG-16 and ResNet-50 Architectures

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Abstract

Recent advancements in computer technology enable people to spend most of the time with the Internet for various purpose. This helps the cybercriminals to commit online crime with a little physical effort than in the real world. The unwanted softwares known as malicious software (malware) are the primary weapon of cybercriminals to conduct cyberattacks. Several approaches have been deployed till date and among all the ML techniques are better than others provided the models are trained with realistic dataset. The Deep Learning (DL) is the future of ML with its success almost in every applications. This research propose a revolutionary architecture grounded in DL that is capable of classifying malware types using a hybrid approaches. The proposed one is a hybrid DL combines VGG-16 and ResNet 50 model that significantly improves accuracy. The performance is verfied through three standard datasets: The Microsoft BIG 2015, Malimg, and MaleVis. The experimental results demonstrate that the proposed method outperforms the most recent techniques with a highest accuracy of 99.28% with the Malimg dataset.

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Peer-review under responsibility of the scientific committee of the International Conference on Machine Learning and Data Engineering

Keywords: Cybersecurity, Malware Detection, Deep Learning, hybrid Deep Learning Model

1. Introduction

The Internet and recent computer technology advancements have made life more pleasant and simpler for individuals [1]. The COVID-19 pandemic has boosted the process of digitization [2]. These advancements also raises the concern about safety and privacy of individuals data and software. The McAfee Covid-19 threat report states that 375 new cybercrimes are reported globally every minute and that malware attacks have surged by 19.02% over the previous four quarters. Market growth for malware analysis is predicted to reach 11.7 billion USD at a compound annual growth rate (CAGR) of 31% between 2019 and 2024. According to a recent estimate by the AV-Test Institute, there are about 3.6 lakh new malwares created every day, or 4.2 malwares every second. According to data spanning the preceding five years, there has been a yearly spike of 100 million emerging malware incidents [3]. The number of

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10.1016/j.procs.2025.04.397

these illicit techniques is frighteningly rising yearly. According to American cyber security firm SonicWall, there will be 5.6 billion attacks and approximately 10 billion pieces of malware discovered in 2020. Malware attacks began as a side project for researchers and hackers and are now a global problem [4].

Malware comes in a variety of forms, such as worms, ransomware, Trojan horses, rootkits, and viruses [5]. Variants of malware have the ability to cause disruptive damage to computer systems, start distributed denial of service (DDoS) harm, and steal private data [1]. Malware is known to impact both business and private systems [6]. Malware was once easily recognized because to its basic coding. However, modern malware writers are making their codes more complex, making it impossible for even the most sophisticated methods to identify them. Unlike traditional malware, which is made to run on the kernel, next-generation malware is harder to spot. Malware of this kind can readily evade security measures such as firewalls and antivirus software. They harm the intended user, spread through numerous extensions, and remain inside a computer system or network indefinitely [7]. Numerous attacks from simple bugs to ransomware and beyond to complex malware such Stuxnet compromises the safety and privacy of individual users and jeopardize the integrity of entire countries [8]. Nevertheless, up until recently, it was employed for terrorist activities that may have resulted in fatalities, as well as financial and commercial losses [5].

In the recent years, Machine Learning (ML) has emerged as the most demanding field of research. This is due to its approach to learn models from variety of data and classify or predict the outcome of newly arrived data. The ML is applied in the field of cybersecutiy up to a greater level for identifying threats. The Deep Learning (DL) techniques have been proven to be more accurate compared to standard ML techniques. Therefore, researchers in the field of malware detection are extensively using DL for their purpose.

DL is extensively applied in numerous fields such as computer vision, image analysis, smart transportation, recognizing human actions, facial expression detection, and NLP [1]. One of such promising application is malware identification. There are methods for identifying malware threats that make use of deep learning technologies, primarily the Convolutional Neural Networks(CNN) [9, 10]. CNN has the ability to classify malware more accurately and quickly than classical malware for a variety of classification situations. CNNs perform exceptionally well when dealing with image and massive data sets.

We propose a hybrid DL model for detection of malware. Precisely, its a crossover model where a CNN is trained for recognition and categorization by rendering each virus as a grayscale image, drawing on prior work. This effort has yielded significant improvement and comparable results with the existing models.

The remaining portion is organized into five sections. Section 2 highlights the importance and features of CNN. Section 3 defines the literature survey of the malware detection using DL. Section 4 presents the proposed DL model for malware detection. Section 5 summarizes the implementation details and results obtain from the experiment. Section 6 enlist the concluding points and future directions.

2. Convolution Neural Network

CNN offers a variety of deep learning models, including multi-layered neural networks, that share a basic structure known as a layer of convolution [11]. A typical architecture of CNN consists of four layers: i) Input, ii) Convolutions, iii) pooling, and iv) fully connected layers. The purpose of the initial convolution layer is to acquire fundamental characteristics such gradient location, color, and edge. The structure of the layers changes as more are added to accommodate higher-level functions, improving our comprehension of the dataset's visuals. The convolutional layer is simply a collection of convolutional filters, or "kernels," which are employed for obtaining different information out of the given image.

Similar to convolution layers, pooling layers assist in reducing the convolution function's spatial size. This reduces the amount of computing required to process the data through dimension reduction. It also helps to extract important rotational and positional elements. In this way, the model training process is effectively continued. Average pooling and Maximum pooling are the two categories of pooling layers. Max Pooling yields the greatest value within the area of the picture which the kernel is occupied when it is applied as a denoiser. The region of the picture occupied by the kernel is used to compute the average values.

The foundation of the CNN structure is a layer called the Fully Connected Layer (FCL). This layer's nodes are connected to all the nodes in all other layers. It functions similarly to a conventional multiple-layer perceptron neural network because it is a feed-forward Artificial Neural Network(ANN) with the flattened feature map. This layer is

Table 1: Summary of literature survey.

Authors	Dataset	Models	Accuracy
[2] by Aslan *et al.*	Malimg	AlexNet + ResNet-50	97.78%.
[5] by Malani *et al.*	Meraz 18	CNN	98.02%
[6] by Bagane *et al.*	Malimg	CNN+BiLSTM	97.4%
[7] by Patil *et al.*	NA	CNN	92.09%
[12] by Vinayakumar *et al.*	NA	CNN+LSTM	98.8%
[13] by Dawra *et al.*	Malimg	CNN+LSTM	98.04%
[15] by Bhardwaj *et al.*	BODMAS	MD-ADA	99.29%
[14] by Al-Khater *et al.*	Malimg	ResNet-18	99.64%

followed by a drop-out layer. The node's ability to generalize is improved by the dropout layer since it inhibits overfitting, an issue that affects all deep learning methods. There are various CNN architectures such as VGG16 and ResNet-50 [12]. By breaking down byte codes, CNN's features for malware detection are recovered. The CNN is used to construct 2-Dimensional Grayscale graphics from malware byte code. Down sampling is used to normalize each sample to a size of 32×32 pixels [3].

3. LITERATURE SURVEY

Here, we provide an overview of a few recent studies on malware detection that use a variety of DL approaches to increase accuracy and are documented in the literature.

Aslan *et al.* [2] have proposed a DL architecture in order to find different malware versions efficiently. The suggested framework employs a hybrid methodology. This method uses a number of comprehensive previously trained networks that incorporate learning through transfer. The first step in gathering malware data involved leveraging multiple large-scale databases. Next, the feature extraction is performed with the help of pre-trained networks. Ultimately, using a supervised learning approach, the deep neural network architecture's training phase is carried out. They have introduced a hybrid model which combines two popular previously trained network models, Alex Net and Resnet - 152 Net. The model is used for evaluating the Microsoft BIG 2015, Malimg, and Malevis datasets, yielding accuracy of 96.78%, 94.88%, and 96.5%, respectively.

In order to facilitate a better analysis, Malani *et al.* [5] have simultaneously trained and validated a variety of DL neural networks, including CNN,GRU, ANN, and CNN+LSTM. Additionally, they have also performed a comparison analysis using a range of performance criteria. According to their analysis, the ANN and GRUs performed poorly, with highest approval percentages of 72.06% and 64.99%, accordingly. The hybrid CNN+LSTM architecture outperformed CNN with a maximum validation accuracy of 97.6%.

Bagane *et al.* [6] have employed hybrid convolutional neural networks, such as CNN+LSTM and CNN+BiLSTM along with CNN, to find patterns in malware sequences. It is reported that the CNN+BiLSTM is found to perform better than CNN and CNN+LSTM.

Dawra *et al.* [13] have visualized that the texture and arrangement of images from the same virus families have similarities. They contrast the CNN-LSTM's accuracy with that of other trained models. They used a DL framework to detect and classify malware by converting the binary malware files into grayscale photos. In comparison to CNN, which only considers spatial features, CNN-LSTM have performed better due to the ability to identify spatial features and temporal as well. The CNN-LSTM model has a 98.04% accuracy during training in class prediction, according to the results. In a similar work by Patil *et al.* [7], CNN on the SMOTE dataset have achieved the increment in accuracy by 2.56% over standard dataset.

Al-Khater *et al.* have used FABEMD technique [14]. The derived images are considered as input videos and put in to 3 dimensional VGG-16 and Resnet-18. Both Malimg and MaleVis datasets are used for training and validation. The classification accuracy over Malimg and MaleVis for VGG-16 are achieved as 96.14% and 98.60% respectively. While for Resnet-18, the accuracy is 99.64% and 99.46% respectively. The overall comparison of works reported in the literature is summarized in Table 1

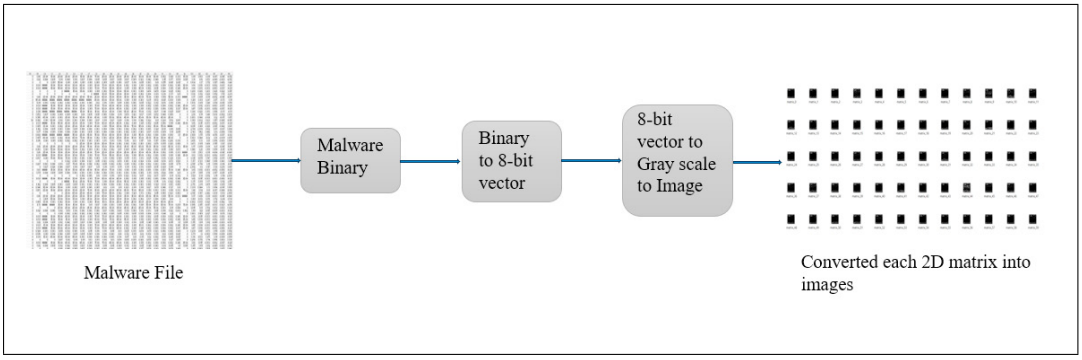


Fig. 1: Preprocessing of malware text file in images

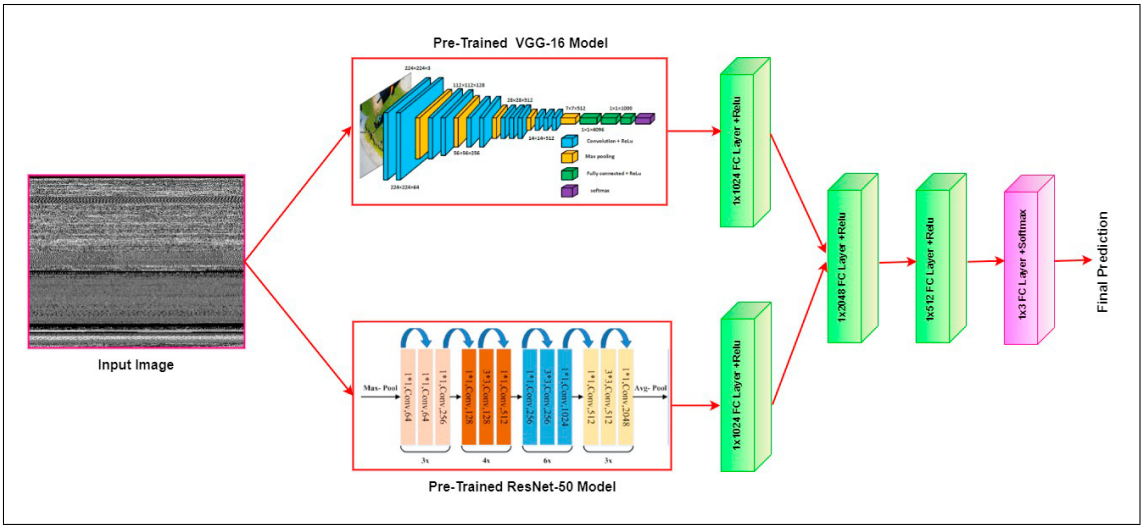


Fig. 2: The Proposed Hybrid Model

4. The Proposed Architecture

The proposed hybrid CNN model to identify malware combines the properties of two pre-trained architectures: the VGG-16 and the ResNet-50. Three fully connected layers are used for prediction with first layer being the fusion of two fully connected layers of size 1×1024. The dataset used in the proposed work consist of image as well as text data. The process adapted to convert the text data into image is provided in the following data preprocessing section.

4.1. Dataset Preprocessing

The data preprocessing steps are aligned Figure 1. Initially, the binary version of the file is fetched and transformed into an unsigned, 8-bit vector. The min-max normalization is used for this purpose. Further, the vectors are transformed to grayscale values that ultimately generates a grayscale image of the file [7].

4.2. Working of proposed architecture

Figure 2 demonstrates the proposed architecture. The input image for the model has dimensions of $256 \times 256 \times 3$. An image is downsampled to its half every time it is provided as input to a layer. However, an image is stopped from being downsampled upto reaching the size of $16 \times 16 \times 3$.

Table 2: Performance Matrix on Microsoft BIG dataset

Family	Precision	Recall	F1-score	Support
Ramanit	0.97	0.99	0.98	310
Lollipop	1.00	0.99	1.00	486
Kelihosver1	1.00	1.00	1.00	633
Kelihosver3	1.00	0.96	0.98	98
Vundo	1.00	1.00	1.00	3
Simda	0.96	0.99	0.98	148
Tracur	0.93	1.00	0.96	77
Obfuscastor.ACY	0.99	0.94	0.96	219
Gatak	1.00	1.00	1.00	200
macro avg	0.98	0.99	0.98	2174
micro avg	0.99	0.99	0.99	2174
samples avg	0.99	0.99	0.99	2174
weighted avg	0.99	0.99	0.99	2174

The fusion of VGG-16 and ResNet-50 is achieved by extracting features from the final max-pooling layer of VGG-16 and the final average-pooling layer of ResNet-50. These features are passed through individual 1×1024 fully connected (FC) layers. The outputs from these layers are then combined into a single 2048-dimensional vector, which merges the features from both models. This fused vector is further refined using additional FC layers before the final classification. This process is further explained as follows. At first, one of the CNN models out of VGG-16 and ResNet-50 is selected and a malware dataset image is fed to it. The resulting features of the max-pooling layer of VGG-16 is then fed to the Fully Connected (FC) layer with ReLU activation function of size 1×1024. Similar, output features of ResNet-50's final average-pooling is provided as input to 1×1024 dimension FC layer associated with ReLU. The output characteristics from the 1×1024 FC layer of VGG-16 and ResNet-50 are fused together to input to the next 1×2048 FC layer. Finally, an additional FC layer of size 1×512 was deployed to extract further essential features. The final classification challenge is completed using last FC layer of size 1×3 with Softmax. The total of 80 % of the dataset images are used for training of the proposed hybrid CNN model and performance is evaluated with the rest of the 20% images.

5. Implementation and Results

Three standard datasets are used for implementing the proposed model. These are The Microsoft BIG 2015 dataset [2], Malimg dataset[16], and MaleVis dataset [17]. The Microsoft BIG 2015 Dataset has 21,741 malware samples from nine distinct classes. Here, each malware sample has a single Hot-Encoding vector for representation. There are total of 9,339 malware date in the form of images categorized to 25 different families and classes in the Malimg Dataset. The image size of this dataset varies. Nevertheless, the size and texture are comparable. The MaleVis Dataset has 9,100 images of malware with 26 distinct families. For convenience, the samples were transformed from 300 x 300 pixels to 224 x 224 pixels. Assessment measures used for evaluation are the standard metrics such as Precision, Recall, ROC, and F1-score. The 80:20 split approach is used to divide the dataset images into training and testing set.

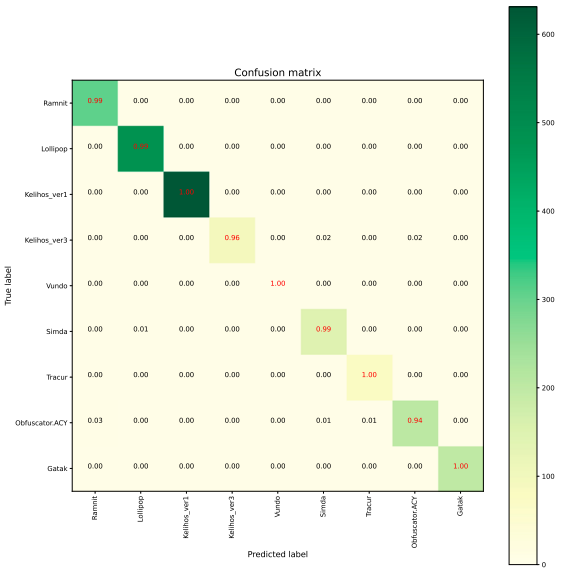
We have used Google Colab environment to implement the hybrid DL model. Table 2, 3, and 4 shows the performance hybrid model on The Microsoft BIG, Malimg, and MaleVis dataset respectively.

The Tables 2, 3, and 4 provide the performance matrices of proposed model over individual malware families and on average data as well. The confusion matrices of the proposed model for all the datasets are shown in Figure 3. Similarly, the ROC curves for all the datasets of the proposed model are depicted by Figure 4.

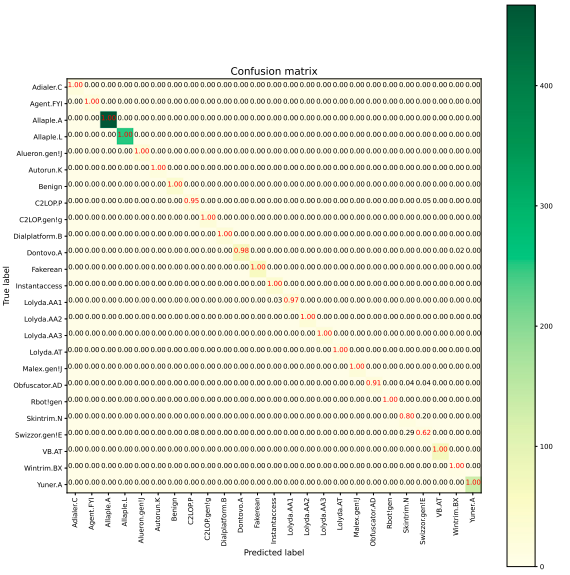
Figure 5a, and 5b display the training accuracy and loss of The Microsoft BIG 2015 dataset respectively over 20 epochs.

Similarly, Figure 6a, and 6b display the training accuracy and loss of Malimg dataset respectively over 20 epochs.

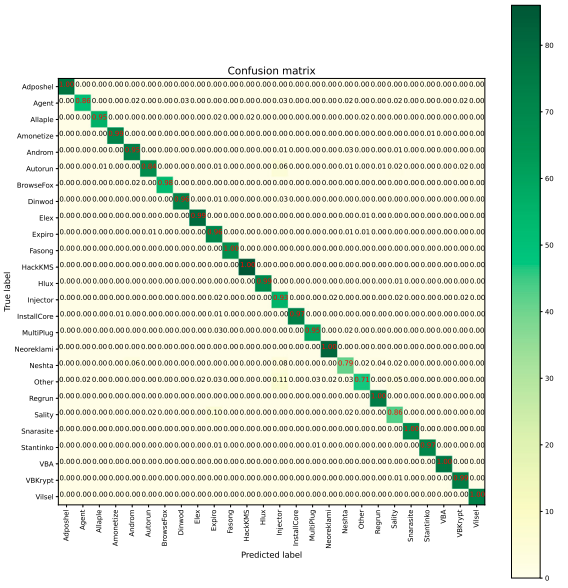
The accuracy comparison with existing models is summarized in Table 5. The work by Aslan *et al.* is the only other work that have implemented the model using the above mentioned three standard datasets of malware. Also,



(a) Confusion matrix of Microsoft BIG Dataset



(b) Confusion matrix of Malimg Dataset.



(c) Confusion matrix of MaleVis Dataset.

Fig. 3: The Confusion Matrix.

the proposed hybrid model obtains the accuracy of 99.5, 99.28, and 98.01 % for the dataset of Microsoft, Maligm and MaleVis dataset respectively. Moreover, the proposed model is able achieve better accuracy that all the existing models for all the trained datasets. The accuracy difference across the datasets can be attributed to their characteristics. The Malimg dataset achieved 99.28% accuracy due to its visually distinct malware samples, while the Microsoft Big Dataset, with 99.15%, is larger and more diverse, presenting more challenges in distinguishing subtle patterns. The MaleVis dataset recorded 98.01% accuracy, due to overlapping features among malware classes, which made detection more difficult for the model.

Table 3: Performance Matrix of Malimg dataset

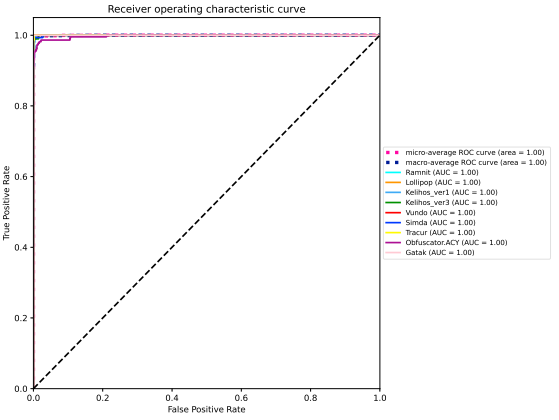
Family	Precision	Recall	F1-score	Support
Autorun.K	1.00	1.00	1.00	17
Agent.FYI	1.00	1.00	1.00	19
Adialer.C	1.00	1.00	1.00	21
Allaple.A	1.00	1.00	1.00	468
C2LOP.P	0.91	0.95	0.93	22
C2LOP.gen!g	1.00	1.00	1.00	39
Lolyda.AA2	1.00	0.97	0.98	30
Dialplatform.B	1.00	1.00	1.00	27
Swizzor.gen!E	0.67	0.80	0.73	20
Dontovo.A	1.00	1.00	1.00	27
Fakerean	1.00	0.98	0.99	66
Allaple.L	1.00	1.00	1.00	251
Instantaccess	1.00	1.00	1.00	62
Lolyda.AA1	0.97	1.00	0.99	35
Alueron.gen!J	1.00	1.00	1.00	28
Lolyda.AT	1.00	1.00	1.00	24
Wintrim.BX	0.91	1.00	0.95	10
Lolyda.AA3	1.00	1.00	1.00	28
Skintrim.N	1.00	1.00	1.00	9
Malex.gen!J	0.95	1.00	0.97	18
Obfuscator.AD	1.00	1.00	1.00	24
VB.AT	1.00	1.00	1.00	69
Rbot!gen	1.00	0.91	0.95	23
Swizzor.gen!I	0.71	0.62	0.67	24
Yuner.A	1.00	1.00	1.00	131
macro avg	0.96	0.97	0.97	1492
micro avg	0.99	0.99	0.99	1492
samples avg	0.99	0.99	0.99	1492
weighted avg	0.99	0.99	0.99	1492

Table 4: Performance Matrix of MaleVis dataset

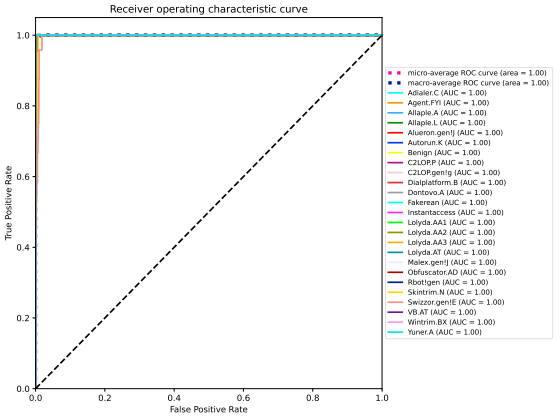
Family	Precision	Recall	F1-score	Support
Adposhel	1.00	1.00	1.00	77
Allaple	0.98	0.95	0.97	59
Agent	0.98	0.86	0.92	59
Amonetize	0.99	0.99	0.99	73
BrowseFox	1.00	0.98	0.99	54
Androm	0.94	0.95	0.94	77
Autorun	0.97	0.84	0.90	81
Elex	0.99	0.99	0.99	81
Dinwod	0.97	0.96	0.97	75
Fasong	1.00	1.00	1.00	67
HackKMS	0.99	1.00	0.99	86
Expiro	0.83	0.96	0.89	74
Hlux	1.00	0.99	0.99	71
MultiPlug	0.95	0.95	0.95	62
InstallCore	1.00	0.97	0.99	75
Injector	0.72	0.93	0.81	59
Neshta	0.82	0.79	0.80	52
Neoreklami	0.98	1.00	0.99	81
Sality	0.80	0.86	0.83	50
Regrun	0.96	1.00	0.98	74
Snarasite	1.00	1.00	1.00	69
VBKrypt	0.95	0.99	0.97	75
VBA	1.00	1.00	1.00	77
Stantinko	0.99	0.97	0.98	73
Vilsel	1.00	1.00	1.00	73
Other	0.92	0.71	0.80	66
macro avg	0.95	0.95	0.95	1820
micro avg	0.95	0.95	0.95	1820
samples avg	0.95	0.95	0.95	1820
weighted avg	0.95	0.95	0.95	1820

6. CONCLUSION

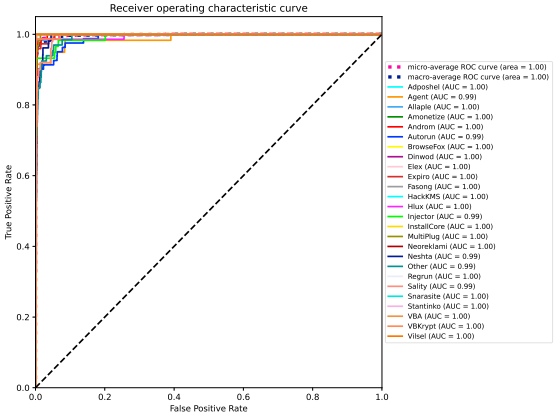
In this paper, A fusion of VGG-16 and ResNet-50 is designed along with multiple dense layers to enhance the malware detection accuracy. Three key datasets are used access the proposed model. The obtained results infers that the proposed model have achieved higher accuracy than existing hybrid models. The proposed model is resistant to obfuscation; it did not use manipulated inputs to identify the adversary's threats. In future we aim to test and enhance the ability of our model to test it on a real-time dataset. Further, we plan to reduce hidden layers of the existing model in order to achieve optimized performance.



(a) ROC Curve of The Microsoft BIG 2015 Dataset

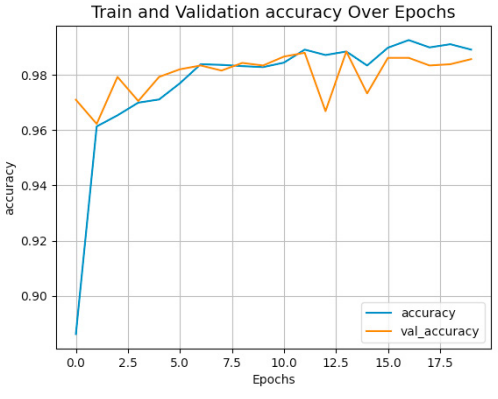


(b) ROC Curve of The Malimg Dataset.

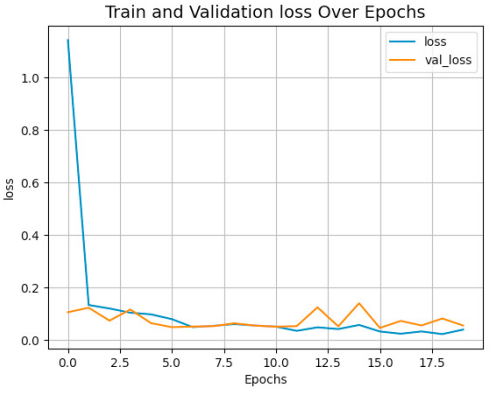


(c) ROC Curve of The MaleVis Dataset.

Fig. 4: The ROC Curve.



(a) The accuracy of The Microsoft BIG 2015 Dataset

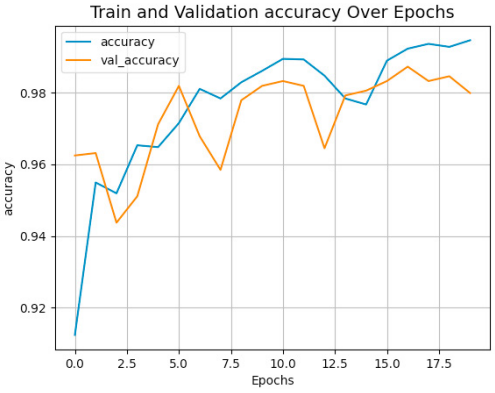


(b) The loss of The Microsoft BIG 2015 Dataset

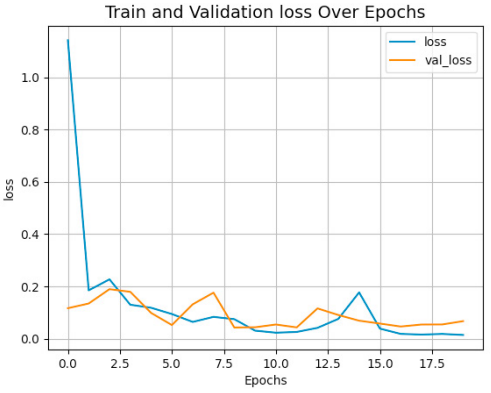
Fig. 5: The accuracy and the loss of The Microsoft BIG 2015 Dataset

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(a) Accuracy over Malimg dataset.



(b) Loss over Malimg dataset.

Fig. 6: The accuracy and loss for Malimg dataset.

Method	Microsoft 2015 dataset	Maligm dataset	MaleVis dataset
Dawra *et al.* [13]	95	-	-
Aslan *et al.* [2]	94.88	97.78	96.5
Nivaashini *et al.* [18]	95.4	-	90.7
Roseline *et al.* [19]	97.2	-	97.53
Bagane *et al.* [6]	-	97.4	-
Patil *et al.* [7]	-	90	-
Vinaya Kumar *et al.* [12]	-	93.6	-
Falana *et al.* [20]	-	-	96.7
Proposed Model	99.15	99.28	98.01

Table 5: Comparison of accuracy with existing detection DL models.

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